

HARDWARE COMPONENTS & SYSTEM ASSEMBLY

Section A — Concept Application

S1

You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

Answer

1. Choose CPU first – e.g. Intel Core i5 (12th Gen+) or AMD Ryzen 5 (5000+), which fixes the required socket and supported chipsets.

2. Select motherboard that:

- **Matches the CPU socket and chipset.**
- **Supports the needed RAM type (DDR4/DDR5), speed, and capacity.**
- **Has required ports (USB, HDMI/DisplayPort, LAN, Wi-Fi).**

3. Pick RAM that:

- **Matches the motherboard's memory type and speed.**
- **Fits within max capacity and slot layout (e.g. 2×8 GB for 16 GB).**
- **16 GB is suitable for typical office multitasking.**

4. Choose storage:

- **NVMe M.2 SSD (e.g. 512 GB) if the board has an M.2 NVMe slot.**
- **Or SATA SSD/HDD if using SATA ports.**

Confirm compatibility by:

- **Checking the motherboard's CPU support list and required BIOS version.**
- **Verifying RAM on the board's QVL/memory compatibility list.**
- **Confirming M.2/SATA support and interface in the motherboard specs.**
- **Cross-checking CPU specs (socket, memory support) on Intel/AMD pages.**

S 2

You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

Answer:

HDD vs. SSD vs. NVMe

For employees whose main work consists of Microsoft Office, email, web applications, document management, spreadsheets, and collaboration tools, the best choice is an NVMe SSD as the primary system drive. It provides fast application launches, low latency, strong multitasking performance, silent operation, and good energy efficiency without the mechanical limitations of an HDD.

HDD vs SSD vs NVMe SSD – Comparison and Recommendation

Feature	HDD (Hard Disk Drive)	SSD (SATA)	NVMe SSD (PCIe)
 Technology	Uses spinning magnetic platters and moving read/write heads.	Uses NAND flash memory with no moving parts. Connects via SATA interface.	Uses NAND flash memory with no moving parts. Connects via PCIe interface using NVMe protocol.
 Speed	Slowest (100–200 MB/s)	Faster (500–560 MB/s)	Fastest (2000–7000+ MB/s)
 Performance	Lower performance, higher latency.	Better performance and lower latency than HDD.	Excellent performance, very low latency.
 Durability & Reliability	Less durable due to moving parts. More prone to damage from shock or vibration.	More durable and reliable because it has no moving parts.	Very reliable and durable, same as SSD.
 Power Consumption	Higher power usage.	Lower power usage.	Lowest power usage among all three.
 Noise	Produces noise.	Silent operation.	Silent operation.
 Capacity	Very high (1TB to 20TB+) at lower cost.	Moderate to high (256GB to 4TB).	Moderate to high (250GB to 4TB+).
 Price	Cheapest per GB.	Higher than HDD.	Highest per GB.
 Best Use	Mass storage, backups, large files where speed is not critical.	Everyday use, OS, applications, office work, general purpose.	High performance tasks, large file transfers, gaming, content creation.

Short Recommendation

- **HDD: Uses spinning magnetic disks. It is inexpensive and offers high capacity, but it is slower, noisier, and more prone to mechanical failure.**

- **SATA SSD: Uses flash memory with no moving parts. It is much faster, quieter, and more reliable than an HDD.**
- **NVMe SSD: A faster type of SSD that uses the PCIe interface and NVMe protocol. It offers lower latency, higher throughput, and better multitasking performance than a SATA SSD.**

S 3

You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

Answer:

GPU and Cooling Roles

GPU

The GPU handles graphics calculations such as 2D/3D rendering, viewport display, shading, animation, and hardware-accelerated effects. This reduces the workload on the CPU and makes design software more responsive, especially when working with complex models, high-resolution textures, or multiple displays. Microsoft notes that CAD, simulation, rendering, and visualization applications can rely heavily on GPU-based 3D rendering.

A suitable workstation GPU should provide:

- Enough VRAM for the size of the design files and textures.**
- Drivers certified or optimized for the design application.**
- Support for the software's graphics API and GPU-acceleration features.**
- Sustained performance rather than only a high short-term boost speed.**

Cooling System

Running design software for several hours keeps the GPU, CPU, voltage regulators, and sometimes memory under continuous load. The cooling system removes this heat using:

- **Heatsinks to absorb heat.**
- **Fans to move warm air out of the case.**
- **Case airflow to supply cool air.**
- **Thermal paste or pads to transfer heat efficiently.**

Good cooling prevents thermal throttling, where the processor or GPU lowers its clock speed to avoid overheating. Intel describes throttling as a protective reduction in clock speed when temperatures exceed safe limits.

In short: the GPU provides the graphics performance, while the cooling system preserves that performance and prevents crashes, instability, and premature component wear during sustained workloads.

S 4

You are assembling a PC from components sourced from three different vendors.

Question: Explain why hardware compatibility matters when assembling a computer, and identify

potential issues that could arise from incompatible components.

Answer:

Potential Issues

- **CPU and motherboard:** The processor socket, chipset, and BIOS must match. An incompatible CPU may not boot at all; for example, Intel processors can require specific chipsets even when the socket appears similar.
- **RAM and motherboard:** The memory type, speed, capacity, and voltage must be supported. Incorrect RAM may cause crashes, failure to boot, or reduced performance.
- **Power supply:** The PSU must provide sufficient wattage and the correct CPU/GPU connectors. An undersized or incompatible PSU can cause shutdowns, instability, or component damage.
- **GPU and motherboard/case:** The graphics card must fit the PCIe slot, case dimensions, and available power connectors. It also requires adequate airflow.

- **Storage:** M.2 drives must match the motherboard's supported interface, such as SATA or NVMe.
- **Cooling:** The CPU cooler must fit the socket and case and have adequate cooling capacity. Poor compatibility can lead to overheating and thermal throttling.
- **Case and motherboard:** The motherboard form factor, such as ATX, microATX, or Mini-ITX, must match the case.

S 5

You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

Answer:

Choosing and Connecting the PSU — Short Version

Choosing the right PSU:

- **Add up total component wattage (CPU, GPU, drives, RAM, etc.) — GPU is usually the biggest draw**
- **Add 20–30% headroom for power spikes and future upgrades**
- **Choose at least 80 PLUS Bronze efficiency (Gold+ preferred for daily heavy use)**
- **Confirm it has the right connectors: 24-pin motherboard, CPU EPS (4+4/8-pin), GPU PCIe power, and SATA power**

Safety checks before powering on:

- **PSU off and unplugged while connecting cables; discharge static first**
- **Confirm 24-pin and CPU power connectors are fully seated (don't confuse EPS with PCIe connectors)**
- **Confirm GPU and SATA power connectors are properly clicked in**
- **Check no cables are pinched or resting on fans**
- **Inspect case interior for loose screws/debris that could cause a short**

- **Confirm motherboard sits on standoffs, not touching the case directly**
- **Verify PSU voltage switch matches your region (if applicable)**

First boot:

- **Power on with the case open (or breadboarded) to watch/listen for issues**
- **Confirm successful POST via debug LEDs, beep codes, or display output**

Bottom line: Correct sizing prevents instability under load; careful connection checks prevent the two most common failures — wrong wattage and shorted/loose connectors.

S 6

You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

Answer:

Confirming POST Success — Short Version

Checks to confirm successful POST:

- **System powers on, fans spin, no unusual noises (grinding, clicking)**
- **Single short beep (if speaker present) or debug LEDs clear after boot**
- **Monitor shows splash/logo screen, then proceeds to BIOS or OS**
- **In BIOS: confirm CPU, RAM (full capacity/speed), and storage drives are all detected correctly**
- **System boots into the OS desktop without errors; no flagged devices in Device Manager**

If POST fails:

1. **Read the diagnostic signal — beep code or debug LED tells you which component (CPU/RAM/GPU/boot) is likely at fault**
2. **Reseat components — RAM (try one stick at a time), GPU, and power connectors**
3. **Breadboard test — strip to CPU + one RAM stick + GPU outside the case to rule out a case short**
4. **Swap parts — substitute known-working components to isolate the fault**

5. Check physical issues — bent socket pins, CPU cooler contact, loose screws/debris

6. RMA the identified faulty component if the issue persists

Bottom line: Confirm POST through visual, audible, and BIOS-level checks before handoff. If it fails, isolate systematically (reseat → breadboard → swap) rather than guessing.

